

# Generation Interconnection System Impact Study Report

## For Queue Project Q16-0590

Clover 138 kV  
January 28, 2017

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0590. The Interconnection Customer proposes a 169.4 MW solar photovoltaic facility interconnecting at the Clover 138 kV substation in the Virginia area. The request was received on March 02, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 317 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,572,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                              | Contingency                                     | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------|-------------------------------------------------|-------------|--------------|----------|
| GLENANNAN to CAL SIERRA DEVELOPMENT 345 KV Line | ROBERTS SW STA SUBSTATION to GLENANNAN 345 KV   | 126.1       | 1,319        | 8.0      |
| CEDAR CREEK to MONROE NE 345 KV Line            | CO LINE RD SUBSTATION to MONROE NE 500 KV       | 120.7       | 1,144        | 7.3      |
| WILKENS to HIGHGROVE 138 KV                     | TABER to DIXLETA 345 KV                         | 120.6       | 794          | 5.4      |
| N POLE to DIXLETA 138 KV                        | DIXLETA to N POLE 138 KV Line 2                 | 119.7       | 815          | 5.4      |
| DIXLETA to NORTH POLE 138 KV Ckt 2              | N PLE to DIXLETA 138 KV Ckt 1                   | 119.7       | 815          | 5.4      |
| WHEELABRATOR BALTIMORE REFUSE to WILKENS 138 KV | TABER to DIXLETA 345 KV                         | 118.8       | 839          | 5.4      |
| ROBERTS SW STA to GLENANNAN 345 KV              | CAL SIERRA DEVELOPMENT to GLENANNAN 345 KV L    | 117.5       | 1,461        | 8.6      |
| NAUVOO to ROBERTS SW STA SUBSTATION 345 KV Line | NAUVOO to CITY WEST DIESEL PLANT 345 KV Line    | 116.7       | 1,505        | 8.4      |
| DRUM to DIXLETA 138 KV                          | TABER to GLENANNAN 345 KV L                     | 115.8       | 1,201        | 5.5      |
| CITY WEST DIESEL PLANT to NESTUCA 345 KV        | NAUVOO SUBSTATION to CITY W DIESEL PLANT 345 KV | 113.5       | 1,127        | 5.2      |
| RIVA ROAD to SHINGLETOWN 345 KV                 | TABER to GLENANNAN 345 KV                       | 110.9       | 1,628        | 6.9      |
| GASTON to MCCARTNEY SUBSTATION 345 KV L         | ROANOKE RAPIDS to MCCARTNEY 345 KV              | 108.5       | 1,322        | 7.3      |
| CHANCELLOR ROW to BEAR ISLAND 345 KV line       | LEAVITT SUBSTATION to BIRCHWOOD 345 KV          | 106.1       | 1,908        | 19.2     |
| BEAR ISLAND to DUNN SIDING 345 KV line          | LEAVITT to BIRCHWOOD 345 KV L                   | 105.4       | 2,285        | 19.4     |

| Monitored Facility                                              | Contingency                                | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------------------------|--------------------------------------------|-------------|--------------|----------|
| BRANDYWINE POWER FACILITY to CAL SIERRA DEVELOPMENT 345 KV line | ROBERTS SW STA to GLENANNAN 345 KV L       | 103.0       | 1,230        | 6.0      |
| CITY WEST DIESEL PLANT to NAUVOO 345 KV Line                    | NAUVOO SUBSTATION to ROBERTS SW STA 345 KV | 102.8       | 1,615        | 7.7      |
| CITY W DIESEL PLANT to ROBERTS SW STA 345 KV                    | NAUVOO to ROBERTS SW STA 345 KV            | 102.4       | 1,249        | 7.2      |
| NESTUCA to POLK CITY 345 KV                                     | NAUVOO to BIRCHWOOD 345 KV                 | 100.7       | 1,259        | 5.4      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                        | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| GLENANNAN to CAL SIERRA DEVELOPMENT 345 KV Line | Rebuild line with higher-capacity conductor | \$37,063,000  | 2% (\$635,000)        |
| CEDAR CREEK to MONROE NE 345 KV Line            | Rebuild line with higher-capacity conductor | \$19,861,000  | 2% (\$336,000)        |
| WILKENS to HIGHGROVE 138 KV                     | Rebuild line with higher-capacity conductor | \$1,157,000   | 2% (\$23,000)         |
| N POLE to DIXLETA 138 KV Line 1                 | Rebuild line with higher-capacity conductor | \$1,406,000   | 2% (\$27,000)         |
| DIXLETA to NORTH POLE 138 KV Ckt 2              | Rebuild line with higher-capacity conductor | \$1,396,000   | 2% (\$26,000)         |
| WHEELABRATOR BALTIMORE REFUSE to WILKENS 138 KV | Rebuild line with higher-capacity conductor | \$1,077,000   | 2% (\$21,000)         |
| ROBERTS SW STA to GLENANNAN 345 KV              | Rebuild line with higher-capacity conductor | \$81,218,000  | 2% (\$1,395,000)      |
| NAUVOO to ROBERTS SW STA SUBSTATION 345 KV Line | Rebuild line with higher-capacity conductor | \$23,761,000  | 2% (\$433,000)        |
| DIXLETA to DRUM 138 KV                          | Rebuild line with higher-capacity conductor | \$3,025,000   | 2% (\$55,000)         |
| CITY WEST DIESEL PLANT to NESTUCA 345 KV        | Rebuild line with higher-capacity conductor | \$37,999,000  | 4% (\$1,324,000)      |
| RIVA ROAD to SHINGLETOWN 345 KV                 | Rebuild line with higher-capacity conductor | \$91,188,000  | 2% (\$1,597,000)      |
| GASTON to MCCARTNEY SUBSTATION 345 KV L         | Reconductor line                            | \$31,705,000  | 1% (\$404,000)        |
| CHANCELLOR ROW to BEAR ISLAND 345 KV            | Reconductor line                            | \$9,162,000   | 2% (\$200,000)        |
| BEAR ISLAND to DUNN SIDING 345 KV Line          | Reconductor line                            | \$16,976,000  | 2% (\$348,000)        |

| Facility                                                        | Required Upgrade           | Cost Estimate | Allocation to Project |
|-----------------------------------------------------------------|----------------------------|---------------|-----------------------|
| BRANDYWINE POWER FACILITY to CAL SIERRA DEVELOPMENT 345 KV line | Reconductor line           | \$11,302,000  | 2% (\$205,000)        |
| NAUVOO to CITY W DIESEL PLANT 345 KV L                          | Reconductor line           | \$13,152,000  | 2% (\$249,000)        |
| CITY W DIESEL PLANT to ROBERTS SW STA 345 KV                    | Reconductor line           | \$3,916,000   | 2% (\$69,000)         |
| NESTUCA to POLK CITY 345 KV                                     | Reconductor line           | \$6,805,000   | 3% (\$225,000)        |
|                                                                 | Total allocated to project |               | \$7,572,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.